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Professional Summary:

- 2 year of hands-on experience as a Data Analyst at Kantar, working on data processing, predictive modelling, forecasting, and analytical reporting for global clients.
- Strong interest in Machine Learning Engineering and Data Engineering, supported by a Post Graduate Diploma in Big Data Analytics (DBDA) from CDAC Bangalore.
- Solid foundation in Machine Learning and Data Engineering concepts, with experience applying them to real-world business and analytics projects.
- Proficient in Python for data manipulation, statistical analysis, modelling, and automation using libraries like Pandas, NumPy, Scikit-learn, and Statsmodels.
- Experienced in Advanced Analytics and Statistical techniques to interpret data trends, identify patterns, and support data-driven decision-making.
- Strong understanding of Data Collection, DBMS principles, SQL queries, and database management for effective data storage and retrieval.
- Hands-on experience with Big Data Technologies, including tools and techniques for processing and analyzing large datasets.
- Skilled in Data Visualization using Excel, Power BI, and Python to create clear, insightful, and business-focused dashboards and reports.
- Practical experience in implementing Machine Learning models for regression, forecasting, and classification to solve real-world analytical problems.

Professional Experience:

KANTAR (DEC 2024 – Present)

Role: Data Analyst

Tools: Python, Excel, Pandas, NumPy, Scikit-learn, Statsmodels

Techniques: Multiple Regression, Random Forest, Forecasting, EDA, Seasonality Analysis, Feature Engineering

Roles & Responsibilities

- Developed end-to-end predictive models to forecast demand for Shell products using 2–3 years of historical data.
- Performed data cleaning & transformation including handling missing values, outliers, and converting weekly data into monthly aggregates.
- Conducted exploratory data analysis (EDA) to identify seasonality, pricing impact, and distribution trends.
- Built and validated multiple regression models to predict 1–2 year future demand.
- Integrated external macroeconomic factors (CPI, GDP, unemployment) and COVID impact to improve model robustness.
- Applied feature engineering techniques including lag variables and trend indicators.
- Delivered client-ready dashboards (Excel) and insight-driven presentations (PPT).
- Collaborated with stakeholders via email to interpret model outputs and implement feedback.
- Ensured model accuracy using statistical validation techniques (R^2 , error metrics, significance testing).
- MROI MODELLING (Marketing ROI Framework)
- Worked on Marketing Mix Modeling (MROI) to analyze drivers of sales including price, distribution, media, and promotions.
- Applied linear and dynamic regression models to measure impact of marketing investments.
- Incorporated adstock transformation to capture carryover effect of media campaigns.
- Modeled diminishing returns (saturation effect) using non-linear transformations.
- Analyzed price elasticity and relative price index (RPI) to understand pricing impact on sales.
- Evaluated media performance using GRP, impressions, and engagement metrics.
- Identified key sales drivers: Distribution (Wtd Distribution), Price, Media investments, Seasonality & macro factors
- Generated insights on ROI optimization and marketing strategy improvement.
- Advanced Analytics & Modeling Work
- Performed correlation, regression, and time-lag analysis to identify relationships between brand perception, awareness, and penetration.
- Built Random Forest and regression models to predict business outcomes and improve decision-making.
- Conducted feature selection and importance analysis to identify key drivers impacting performance.
- Created scatter plots and visualizations using Python to analyze relationships between variables and support insights.
- Applied EDA techniques to understand trends, seasonality, and customer behavior patterns.
- Demand Power & Predictive Modeling
- Developed Random Forest models to estimate demand power using key features like familiarity, consideration, and imagery.
- Performed data preprocessing and encoding (ordinal + binary encoding) to prepare structured datasets for modeling.

- Engineered relative index-based features to normalize and compare performance across categories.
- Achieved strong model performance (high R^2 score) and identified key influencing factors through feature importance.
- Generated actionable insights by comparing model outputs and identifying gaps in performance.
- Excel & Pivot Skills Used - Expert in Pivot Tables, Pivot Charts, dashboards, and data summaries.
- Strong with formulas: VLOOKUP, XLOOKUP, INDEX-MATCH, SUMIFS, COUNTIFS, IF/IFS, DATE, ROUND.
- Skilled in Power Query for fast data cleaning and transformation.
- Good at What-If Analysis, Goal Seek, and automated Excel reporting.

WBL INTERN IN AI/ML AND PYTHON:

September 2023 – March 2024

National Institute of Electronics and Information Technology, Muzaffarpur.

- Gained foundational knowledge of Python and machine learning through self-study.
- Learned key AI/ML concepts like supervised/unsupervised learning, model evaluation, and data preprocessing.
- Built machine learning projects using algorithms such as Linear Regression, K-Means, and Decision Trees.
- Tools Used: Python, Machine Learning, HTML, CSS.

Web Development Intern: July 2022 – August 2022 • IIT Bhubaneswar

- Developed a web app for users to view the current weather and forecast.
- Knowledge in various web development technologies including HTML, CSS, JavaScript, Node.js, and SQL.

Skills:

Programming languages	: Python, Java, R, SQL
Machine Learning	: Linear Regression, Logistic Regression, Decision Trees, Random Forest, SVM, K-Nearest Neighbors (KNN), PCA, Neural Networks, etc.
Data Manipulation	: Pandas, NumPy
Data Visualization	: Tableau, Matplotlib, Seaborn
Databases	: MySQL, MongoDB
Big Data Technologies	: PySpark, Kafka, Hadoop
Tools	: Jupyter Notebooks, Excel, Google Colab, Microsoft Visual Studio Code, Git
Web Development	: HTML, CSS

Personal Project:

DGA Detection Using Deep Learning

- Currently working on developing a deep learning model to detect Domain Generation Algorithms (DGAs) used by malware for command-and-control (C2) servers.
- Developed a deep learning model utilizing recurrent neural networks (RNNs) with LSTM layers for accurate identification of DGA-generated domain names.
- Designed and trained an LSTM-based model to distinguish between benign and malicious domains, achieving high accuracy and low false-positive rates.
- Implemented model evaluation metrics such as precision, recall, F1-score, and AUC-ROC curve.
- Contributed to improved network security and malware detection by providing an automated and efficient solution for identifying and mitigating DGA threats.
- Technologies Used: Python, TensorFlow/Keras, LSTM networks

Breast Cancer Detection Using Machine Learning Classifiers

- Designed a machine learning model to classify breast cancer tumors based on patient data.
- Evaluated multiple classifiers; XGBoost showed the highest accuracy (98.24%) after tuning.
- Measured performance using precision, recall, F1-score, and ROC-AUC.
- Deployed the final model for real-time tumor classification.
- Technologies: Python, Scikit-learn, XGBoost, Random Forest, Logistic Regression

Education:

- PG Diploma in Big Data Analytics (PG-DBDA):
- Centre for Development of Advanced Computing (CDAC), Bangalore March 2024 – August 2024
- Percentage: 70%
- Bachelor of Technology (B. Tech), Computer Science and Engineering Sitamarhi Institute of Technology
- July 2020 – 2023
- CGPA: 8.45
- Diploma in Computer Science and Engineering Government Polytechnic Muzaffarpur July 2017 – 2020
- CGPA: 8.37

Certifications:

- SQL Certification from Hacker Rank
- Python Certification from Hacker Rank
- SQL Certification from Spoken Tutorial IIT Bombay
- Internet of Things Certification from NPTEL